

SIMPLE COMMANDS:

TOPIC 2 : Directory Management

1. ls

```
Last login: Wed Aug 12 17:52:31 on ttys007
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_api       Documents          Music              Public
abc.c              bookflow_fastapi   Downloads          mysql              PyCharmMiscProject
abc.c#y            CREATE             dsa 3 odd sem     OSSP              PycharmProjects
Applications       data               Library            package-lock.json  show
blockhack hackathon Desktop            Movies             Pictures
amrutha@Amruthas-MacBook-Air ~ %
```

2. ls-l

```
amrutha@Amruthas-MacBook-Air ~ % ls -l
total 144
-rwxr-xr-x  1 amrutha  staff  33424 Jul 31 13:54 abc
-rw-r--r--  1 amrutha  staff    80 Jul 31 13:51 abc.c
-rw-r--r--  1 amrutha  staff    81 Jul 29 12:39 abc.c#y
drwx-----@ 5 amrutha  staff   160 Jun 19 10:23 Applications
drwxr-xr-x  2 amrutha  staff    64 Aug  7 09:39 blockhack hackathon
drwxr-xr-x  3 amrutha  staff    96 Aug 12 15:31 bookflow_api
drwxr-xr-x  5 amrutha  staff   160 Aug 12 17:20 bookflow_fastapi
-rw-r--r--  1 amrutha  staff  17827 Jul 30 11:02 CREATE
drwxr-xr-x  3 amrutha  staff    96 Aug  3 14:53 data
drwx-----@ 138 amrutha  staff   4416 Aug 12 17:59 Desktop
drwx-----@  5 amrutha  staff   160 Aug 11 13:52 Documents
drwx-----@ 337 amrutha  staff  10784 Aug 12 18:09 Downloads
drwxr-xr-x  3 amrutha  staff    96 Jul 23 12:20 dsa 3 odd sem
drwx-----@ 91 amrutha  staff   2912 Jul 25 08:50 Library
drwx-----  5 amrutha  staff   160 Apr 22 21:03 Movies
drwx-----+ 4 amrutha  staff   128 Mar 20 10:47 Music
-rw-r--r--  1 amrutha  staff    51 Jul 30 11:02 mysql
drwxr-xr-x 17 amrutha  staff   544 Aug  3 12:41 OSSP
-rw-r--r--@  1 amrutha  staff    86 May 11 10:41 package-lock.json
drwx-----+ 6 amrutha  staff   192 Jun 13 20:31 Pictures
drwxr-xr-x+ 4 amrutha  staff   128 Feb 21 12:34 Public
drwxr-xr-x@ 4 amrutha  staff   128 Apr 16 14:18 PyCharmMiscProject
drwxr-xr-x@ 3 amrutha  staff    96 Apr 16 14:18 PycharmProjects
-rw-r--r--  1 amrutha  staff     0 Jul 30 11:03 show
amrutha@Amruthas-MacBook-Air ~ %
```

3. ls -a

```
amrutha@Amruthas-MacBook-Air ~ % ls -a
.          .DS_Store          .npmrc          abc.c          Movies
..         .gemini           .oracle_jre_usage abc.c#y        Music
.antigravity .gitconfig       .psql_history   Applications   mysql
.antigravity-ide .homebrew       .redhat         blockhack hackathon OSSP
.anydesk     .idlerc         .ssh            bookflow_api   package-lock.json
.arduinoIDE  .ipython        .Trash          bookflow_fastapi Pictures
.bash_history .junie          .vscode         CREATE         Public
.cache       .Library        .vscode-shared  data          PyCharmMiscProject
.CFUserTextEncoding .lima          .zprofile       Desktop       PyCharmProjects
.claude     .local         .zprofile.pysave Documents      show
.codex      .mongodb       .zsh_history    Downloads
.config     .mysql_history .zsh_sessions  dsa 3 odd sem
.copilot    .npm           abc             Library
```

4. cd

```
amrutha@Amruthas-MacBook-Air ~ % pwd
/Users/amrutha
amrutha@Amruthas-MacBook-Air ~ % cd ~/Documents
amrutha@Amruthas-MacBook-Air Documents % pwd
/Users/amrutha/Documents
```

5. cd ..

```
amrutha@Amruthas-MacBook-Air Documents % cd ..
amrutha@Amruthas-MacBook-Air ~ % pwd
/Users/amrutha
amrutha@Amruthas-MacBook-Air ~ %
```

6. mkdir

```
amrutha@Amruthas-MacBook-Air ~ % mkdir testfolder
amrutha@Amruthas-MacBook-Air ~ % ls
abc          bookflow_api      Documents      Music          Public
abc.c        bookflow_fastapi  Downloads     mysql          PyCharmMiscProject
abc.c#y      CREATE           dsa 3 odd sem OSSP           PyCharmProjects
Applications data             Library       package-lock.json show
blockhack hackathon Desktop         Movies        Pictures       testfolder
```

7. rmdir

```
amrutha@Amruthas-MacBook-Air ~ % rmdir testfolder
amrutha@Amruthas-MacBook-Air ~ % ls
abc          bookflow_api      Documents      Music          Public
abc.c        bookflow_fastapi  Downloads     mysql          PyCharmMiscProject
abc.c#y      CREATE           dsa 3 odd sem OSSP           PyCharmProjects
Applications data             Library       package-lock.json show
blockhack hackathon Desktop         Movies        Pictures
```

8. find

```
amrutha@Amruthas-MacBook-Air ~ % mkdir -p ~/os_test
amrutha@Amruthas-MacBook-Air ~ % touch ~/os_test/sample.txt
amrutha@Amruthas-MacBook-Air ~ % find ~ -name "sample.txt"
find: /Users/amrutha/Music/Music: Operation not permitted
find: /Users/amrutha/Pictures/Photos Library.photoslibrary: Operation not permitted
/Users/amrutha/Desktop/TextHack/sample.txt
find: /Users/amrutha/Library/Application Support/CallHistoryTransactions: Operation not permitted
find: /Users/amrutha/Library/Application Support/CloudDocs: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.sharedfilelist: Operation not permitted
find: /Users/amrutha/Library/Application Support/Knowledge: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.TCC: Operation not permitted
find: /Users/amrutha/Library/Application Support/FileProvider: Operation not permitted
find: /Users/amrutha/Library/Application Support/AddressBook: Operation not permitted
find: /Users/amrutha/Library/Application Support/FaceTime: Operation not permitted
find: /Users/amrutha/Library/Application Support/DifferentialPrivacy: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.avfoundation/Frecents: Operation not permitted
find: /Users/amrutha/Library/Application Support/CallHistoryDB: Operation not permitted
find: /Users/amrutha/Library/Assistant/SiriVocabulary: Operation not permitted
find: /Users/amrutha/Library/Daemon Containers: Operation not permitted
find: /Users/amrutha/Library/Autosave Information: Operation not permitted
```

9. locate

```
amrutha@Amruthas-MacBook-Air ~ % locate sample.txt

WARNING: The locate database (/var/db/locate.database) does not exist.
To create the database, run the following command:

    sudo launchctl load -w /System/Library/LaunchDaemons/com.apple.locate.plist

Please be aware that the database can take some time to generate; once
the database has been created, this message will no longer appear.

amrutha@Amruthas-MacBook-Air ~ % mdfind "sample.txt"
2026-08-12 18:52:35.453 mdfind[50593:9947903] [UserQueryParser] Loading keywords and predicates for locale "en_US"
2026-08-12 18:52:35.453 mdfind[50593:9947903] [UserQueryParser] Loading keywords and predicates for locale "en"
/Users/amrutha/os_test/sample.txt
```

TOPIC 3 : File Management

1. touch

```
amrutha@Amruthas-MacBook-Air ~ % touch
usage: touch [-A [-l[hh]mm]SS] [-achm] [-r file] [-t [[CC]YY]MMDDhhmm[.SS]]
        [-d YYYY-MM-DDThh:mm:SS[.frac][tz]] file ...
amrutha@Amruthas-MacBook-Air ~ % touch test.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    dsa 3 odd sem      OSSP               show
abc.c              CREATE              Library            package-lock.json  test.txt
abc.c#y            data                Movies             Pictures
Applications        Desktop              Music              Public
blockhack hackathon Documents            mysql              PyCharmMiscProject
bookflow_api        Downloads            os_test            PycharmProjects
amrutha@Amruthas-MacBook-Air ~ %
```

2. cat

```
amrutha@Amruthas-MacBook-Air ~ % echo "Hello from my Mac" > test.txt
amrutha@Amruthas-MacBook-Air ~ % cat test.txt
Hello from my Mac
amrutha@Amruthas-MacBook-Air ~ % cp test.txt copy.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    Downloads           os_test             PycharmProjects
abc.c              copy.txt            dsa 3 odd sem      OSSP                show
abc.c#y            CREATE              Library            package-lock.json  test.txt
Applications        data                Movies             Pictures
Applications        Desktop              Music              Public
blockhack hackathon Documents            mysql              PyCharmMiscProject
bookflow_api        Downloads
amrutha@Amruthas-MacBook-Air ~ %
```

3. cp

```
amrutha@Amruthas-MacBook-Air ~ % cp test.txt copy.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    Downloads          os_test            PycharmProjects
abc.c              copy.txt            dsa 3 odd sem     OSSP              show
abc.c#y            CREATE              Library            package-lock.json  test.txt
Applications       data                Movies             Pictures
blockhack hackathon Desktop             Music              Public
bookflow_api       Documents           mysql              PyCharmMiscProject
amrutha@Amruthas-MacBook-Air ~ %
```

4. mv

```
amrutha@Amruthas-MacBook-Air ~ % mv copy.txt renamed.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    dsa 3 odd sem     OSSP              renamed.txt
abc.c              CREATE              Library            package-lock.json  show
abc.c#y            data                Movies             Pictures            test.txt
Applications       Desktop             Music              Public
blockhack hackathon Documents           mysql              PyCharmMiscProject
bookflow_api       Downloads           os_test            PycharmProjects
amrutha@Amruthas-MacBook-Air ~ %
```

5. rm

```
amrutha@Amruthas-MacBook-Air ~ % rm renamed.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    dsa 3 odd sem     OSSP              show
abc.c              CREATE              Library            package-lock.json  test.txt
abc.c#y            data                Movies             Pictures
Applications       Desktop             Music              Public
blockhack hackathon Documents           mysql              PyCharmMiscProject
bookflow_api       Downloads           os_test            PycharmProjects
amrutha@Amruthas-MacBook-Air ~ %
```

6. head

```
amrutha@Amruthas-MacBook-Air ~ % head test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
amrutha@Amruthas-MacBook-Air ~ % head -5 test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
amrutha@Amruthas-MacBook-Air ~ %
```

7.tail

```
[amrutha@Amruthas-MacBook-Air ~ % tail test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
[amrutha@Amruthas-MacBook-Air ~ % tail -5 test.txt
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
amrutha@Amruthas-MacBook-Air ~ % █
```

8.more

```
amrutha@Amruthas-MacBook-Air ~ % more test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
```

9. less

```
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
```

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~

~

(END)

10. wc

```
Last login: Wed Aug 12 19:23:32 on ttys007
amrutha@Amruthas-MacBook-Air ~ % wc test.txt
  10      30     159 test.txt
amrutha@Amruthas-MacBook-Air ~ % wc -l test.txt
  10 test.txt
amrutha@Amruthas-MacBook-Air ~ % wc -w test.txt
  30 test.txt
amrutha@Amruthas-MacBook-Air ~ % wc -c test.txt
 159 test.txt
amrutha@Amruthas-MacBook-Air ~ %
```

11. diff

```
[amrutha@Amruthas-MacBook-Air ~ % cp test.txt test2.txt
[amrutha@Amruthas-MacBook-Air ~ % echo "Line 11: Kiwi" >> test2.txt
[amrutha@Amruthas-MacBook-Air ~ % diff test.txt test2.txt
10a11
> Line 11: Kiwi
amrutha@Amruthas-MacBook-Air ~ %
```

12. sort

```
amrutha@Amruthas-MacBook-Air ~ % echo "Mango
Apple
Banana
Apple
Orange
[Banana" > names.txt
[amrutha@Amruthas-MacBook-Air ~ % cat names.txt
Mango
Apple
Banana
Apple
Orange
Banana
[amrutha@Amruthas-MacBook-Air ~ % sort names.txt
Apple
Apple
Banana
Banana
Mango
Orange
amrutha@Amruthas-MacBook-Air ~ %
```

13. uniq

```
[amrutha@Amruthas-MacBook-Air ~ % sort names.txt | uniq
Apple
Banana
Mango
Orange
```

14. grep

```
amrutha@Amruthas-MacBook-Air ~ % cat names.txt
Mango
Apple
Banana
Apple
Orange
Banana
amrutha@Amruthas-MacBook-Air ~ % grep "Apple" names.txt
Apple
Apple
amrutha@Amruthas-MacBook-Air ~ %
```

TOPIC 4 : File Editing

1. nano

```
amrutha@Amruthas-MacBook-Air ~ % nano editor_test.txt
amrutha@Amruthas-MacBook-Air ~ % cat editor_test.txt
Hello, this file was created using Nano.
amrutha@Amruthas-MacBook-Air ~ %
```

2. vi

```
Last login: Wed Aug 12 19:26:46 on ttys007
amrutha@Amruthas-MacBook-Air ~ % vi editor_test.txt
amrutha@Amruthas-MacBook-Air ~ % cat editor_test.txt
Hello, this file was created using Nano.
amrutha@Amruthas-MacBook-Air ~ %
```

TOPIC 5 : C Program Compilation and Execution

1. gcc

```
amrutha@Amruthas-MacBook-Air ~ % nano hello.c
amrutha@Amruthas-MacBook-Air ~ % gcc hello.c -o hello
amrutha@Amruthas-MacBook-Air ~ %
```

2. ./program

```
amrutha@Amruthas-MacBook-Air ~ % ./hello
Hello from C program!
amrutha@Amruthas-MacBook-Air ~ %
```

3. which

```
amrutha@Amruthas-MacBook-Air ~ % which gcc
/usr/bin/gcc
```

4. whereis

```
amrutha@Amruthas-MacBook-Air ~ % whereis
usage: whereis [-abmqu] [-BM dir ... -f] program ...
amrutha@Amruthas-MacBook-Air ~ % whereis gcc
gcc: /usr/bin/gcc
amrutha@Amruthas-MacBook-Air ~ % which
amrutha@Amruthas-MacBook-Air ~ %
```

5. man

```
CLANG(1)                                Clang                                CLANG(1)

NAME
    clang - the Clang C, C++, and Objective-C compiler

SYNOPSIS
    clang [options] filename ...

DESCRIPTION
    clang is a C, C++, and Objective-C compiler which encompasses preprocessing, parsing, optimization, code generation, assembly, and linking. Depending on which high-level mode setting is passed, Clang will stop before doing a full link. While Clang is highly integrated, it is important to understand the stages of compilation, to understand how to invoke it. These stages are:

    Driver The clang executable is actually a small driver which controls the overall execution of other tools such as the compiler, assembler and linker. Typically you do not need to interact with the driver, but you transparently use it to run the other tools.

    Preprocessing
        This stage handles tokenization of the input source file, macro expansion, #include expansion and handling of other preprocessor directives. The output of this stage is typically called a ".i" (for C), ".ii" (for C++), ".mi" (for Objective-C), or ".mii" (for Objective-C++) file.

    Parsing and Semantic Analysis
        This stage parses the input file, translating preprocessor tokens into a parse tree. Once in the form of a parse tree, it applies semantic analysis to compute types for expressions as well and determine whether the code is well formed. This stage is responsible for generating most of the compiler warnings as well as parse errors. The output of this stage is an "Abstract Syntax Tree" (AST).
```

TOPIC 6 : Process Management

1. ps

```
amrutha@Amruthas-MacBook-Air ~ % ps
  PID TTY          TIME CMD
 60296 ttys007    0:00.02 -zsh
amrutha@Amruthas-MacBook-Air ~ %
```

2. ps -ef

```
amrutha@Amruthas-MacBook-Air ~ % ps -ef
UID      PID  PPID  C  STIME  TTY          TIME CMD
0         1      0  0  16Jul26 ??        77:09.02 /sbin/launchd
0        511      1  0  16Jul26 ??        35:04.19 /usr/libexec/logd
0        513      1  0  16Jul26 ??        1:52.77 /usr/libexec/UserEventAgent (System)
0        515      1  0  16Jul26 ??       23:44.05 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/FSEvents.framework/Versions/A/Support/fseventsd
0        516      1  0  16Jul26 ??        1:30.71 /System/Library/PrivateFrameworks/MediaRemote.framework/Support/mediare
moted
0        519      1  0  16Jul26 ??        0:45.03 /usr/sbin/systemstats --daemon
0        523      1  0  16Jul26 ??        7:08.94 /usr/libexec/configd
0        525      1  0  16Jul26 ??        5:01.31 /System/Library/CoreServices/powerd.bundle/powerd
0        526      1  0  16Jul26 ??        0:00.03 /usr/libexec/IOMFB_bics_daemon
0        530      1  0  16Jul26 ??        0:00.39 /usr/libexec/remoted
306      534      1  0  16Jul26 ??        0:00.03 /System/Library/PrivateFrameworks/CoreSpeech.framework/corespeechd_syst
em
0        536      1  0  16Jul26 ??        0:13.09 /usr/libexec/watchdogd
0        540      1  0  16Jul26 ??       28:57.18 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mds
0        542      1  0  16Jul26 ??        0:01.33 /usr/libexec/kernelmanagerd
0        543      1  0  16Jul26 ??        4:39.14 /usr/libexec/diskarbitrationd
0        546      1  0  16Jul26 ??        4:50.82 /usr/libexec/coreduetd
0        547      1  0  16Jul26 ??        0:52.38 /usr/sbin/syslogd
0        549      1  0  16Jul26 ??        0:53.66 /usr/libexec/thermalmonitord
0        550      1  0  16Jul26 ??        4:25.35 /System/Library/PrivateFrameworks/CoreDuetContext.framework/Resources/c
ontextstored
0        551      1  0  16Jul26 ??        4:00.00 /usr/libexec/xprotectd
0        552      1  0  16Jul26 ??       27:35.38 /usr/libexec/opendirectoryd
0        553      1  0  16Jul26 ??        2:33.68 /System/Library/PrivateFrameworks/ApplePushService.framework/apsd
0        554      1  0  16Jul26 ??       58:23.08 /System/Library/CoreServices/launchservicesd
```

3. top

```
Processes: 819 total, 3 running, 816 sleeping, 4404 threads
Load Avg: 2.66, 2.97, 3.12 CPU usage: 16.62% user, 6.14% sys, 77.23% idle
SharedLibs: 642M resident, 127M data, 93M linkedit. MemRegions: 0 total, 0B resident, 0B private, 1646M shared.
PhysMem: 15G used (2476M wired, 4767M compressor), 107M unused.
VM: 370T vsize, 6144M framework vsize, 4471(0) swapins, 75960(0) swapouts.
Networks: packets: 211970048/93G in, 21412992/15G out. Disks: 35201957/790G read, 20322694/364G written.
19:59:21

PID    COMMAND      %CPU  TIME    #TH   #WQ    #PORTS MEM    PURG    CMPRS    PGRP  PPID  STATE  BOOSTS      %CPU_ME
587    WindowServer 51.4   18:11:48 24    5     10396+ 710M-  16M-    252M   587   1     sleeping *0[1]      1.43899
48745  Google Chrom 35.8   29:18:03 42/1   1     561-   599M+  0B      250M-  96762 96762 running *0[7]      0.00000
32277  promceefplug 24.4   10:45:01 14     5     252    228M  0B      49M    32270 32270 sleeping *1[5]      0.08337
32270  wpsoffice    20.6   09:19:09 82     1     1741+  1327M  112K    896M   32270 1     sleeping *1[16416] 11.7587
85893  promceefplug 14.0   31:46:04 22     1     253    47M+   0B      21M    32270 32270 sleeping *0[1]      0.00000
593    coreaudioid 13.5   85:09:04 14     2     8670   40M    0B      25M    593   1     sleeping *0[1]      0.15385
0      kernel_task  11.0   13:28:37 622/11 0     0       15M    0B      0B     0     0     running  0[0]      0.00000
96893  Google Chrom 9.8    19:00:02 19     1     271    28M    0B      18M    96762 96762 sleeping *0[13701] 0.03369
61795  top          9.2    00:01:95 1/1    0     30     10M    0B      0B     61795 60296 running *0[1]      0.00000
96803  Google Chrom 7.1    01:54:49 25     3     502    608M+  32M-    136M   96762 96762 sleeping *1[31603] 0.76359
96762  Google Chrom 5.3    59:56:32 46     2     1851   381M+  0B      184M-  96762 1     sleeping *4[4295] 0.04954
87577  Google Chrom 4.0    15:36:65 34     1     441    340M-  0B      165M-  96762 96762 sleeping 0[81]    0.00000
41981  wpsbackend   2.1    34:25:78 29     1     243    48M    0B      37M    41981 32270 sleeping *0[3468] 0.00000
81723  WhatsApp     2.1    03:01:39 36     5     3065-  1269M- 1440K   792M- 81723 1     sleeping *1846[31] 0.00000
470    con.apple.Ds 1.0    01:55:00 7      5     1405-  32M-   0B      4704K  470   1     sleeping 0[1]      0.00000
```

4. kill

```
amrutha@Amruthas-MacBook-Air ~ % sleep 1000 &
[2] 62597
[3] 62757
zsh: no matches found: [2]
amrutha@Amruthas-MacBook-Air ~ % ps -p 62597
  PID TTY          TIME CMD
 62597 ttys007    0:00.00 sleep 1000
amrutha@Amruthas-MacBook-Air ~ % kill 62597
amrutha@Amruthas-MacBook-Air ~ %
[2] - terminated  sleep 1000
amrutha@Amruthas-MacBook-Air ~ % ps -p 62597
  PID TTY          TIME CMD
```

5. kill -9

```
amrutha@Amruthas-MacBook-Air ~ % sleep 1000 &
[2] 63098
amrutha@Amruthas-MacBook-Air ~ % ps -p 63025
  PID TTY          TIME CMD
amrutha@Amruthas-MacBook-Air ~ %
amrutha@Amruthas-MacBook-Air ~ % kill -9 63025
kill: kill 63025 failed: no such process
amrutha@Amruthas-MacBook-Air ~ % ps -p 63025
  PID TTY          TIME CMD
amrutha@Amruthas-MacBook-Air ~ %
```

Topic 7: Job Control

1. jobs

```
amrutha@Amruthas-MacBook-Air ~ % jobs
[1]  running    sleep 1000
[2]  running    sleep 1000
[3] - running    sleep 1000
[4] + running    sleep 1000
amrutha@Amruthas-MacBook-Air ~ %
```

2. bg

```
[amrutha@Amruthas-MacBook-Air ~ % sleep 1000
^Z
zsh: suspended  sleep 1000
[amrutha@Amruthas-MacBook-Air ~ % bg
[5] - continued  sleep 1000
[amrutha@Amruthas-MacBook-Air ~ % jobs
[1]   running    sleep 1000
[2]   running    sleep 1000
[3]   running    sleep 1000
[4] + running    sleep 1000
[5] - running    sleep 1000
amrutha@Amruthas-MacBook-Air ~ %
```

3. fg

```
[amrutha@Amruthas-MacBook-Air ~ % fg
[4] - running    sleep 1000
^C
amrutha@Amruthas-MacBook-Air ~ %
```

4. &

```
amrutha@Amruthas-MacBook-Air ~ % sleep 1000 &
[1] 62430
```

Topic 8: File Permissions and Links

1. chmod

```
[amrutha@Amruthas-MacBook-Air ~ % touch permission.txt
[amrutha@Amruthas-MacBook-Air ~ % ls -l permission.txt
-rw-r--r--  1 amrutha  staff   0 Aug 12 20:15 permission.txt
[amrutha@Amruthas-MacBook-Air ~ % chmod 644 permission.txt
[amrutha@Amruthas-MacBook-Air ~ % ls -l permission.txt
-rw-r--r--  1 amrutha  staff   0 Aug 12 20:15 permission.txt
amrutha@Amruthas-MacBook-Air ~ %
```

2. chmod 755

```
amrutha@Amruthas-MacBook-Air ~ % chmod 755 permission.txt
amrutha@Amruthas-MacBook-Air ~ % ls -l permission.txt
-rwxr-xr-x 1 amrutha staff 0 Aug 12 20:15 permission.txt
amrutha@Amruthas-MacBook-Air ~ %
```

3. ln

```
amrutha@Amruthas-MacBook-Air ~ % echo "Hello from original file" > original.txt
amrutha@Amruthas-MacBook-Air ~ %
[1] done sleep 1000
amrutha@Amruthas-MacBook-Air ~ % ln original.txt hardlink.txt
amrutha@Amruthas-MacBook-Air ~ % ls -l original.txt hardlink.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 hardlink.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 original.txt
amrutha@Amruthas-MacBook-Air ~ % cat hardlink.txt
Hello from original file
amrutha@Amruthas-MacBook-Air ~ %
```

4. ln -s

```
amrutha@Amruthas-MacBook-Air ~ % ln -s original.txt symlink.txt
amrutha@Amruthas-MacBook-Air ~ % ls -l symlink.txt
lrwxr-xr-x 1 amrutha staff 12 Aug 12 20:20 symlink.txt -> original.txt
amrutha@Amruthas-MacBook-Air ~ % cat symlink.txt
Hello from original file
amrutha@Amruthas-MacBook-Air ~ %
```

Topic 9: Disk and Memory Information

1. df -h

```
amrutha@Amruthas-MacBook-Air ~ % df -h
Filesystem      Size  Used Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1  460Gi  12Gi  318Gi    4%   459k  3.3G    0%      /
devfs           213Ki  213Ki   0Bi  100%    736    0  100%    /dev
/dev/disk3s6    460Gi  2.0Gi  318Gi    1%     2  3.3G    0%    /System/Volumes/VM
/dev/disk3s2    460Gi  8.4Gi  318Gi    3%   1.4k  3.3G    0%    /System/Volumes/Preboot
/dev/disk3s4    460Gi  2.4Mi  318Gi    1%    65  3.3G    0%    /System/Volumes/Update
/dev/disk1s2    500Mi  6.0Mi  482Mi    2%     1  4.9M    0%    /System/Volumes/xarts
/dev/disk1s1    500Mi  6.0Mi  482Mi    2%    33  4.9M    0%    /System/Volumes/iSCPreboot
/dev/disk1s3    500Mi  1.4Mi  482Mi    1%    94  4.9M    0%    /System/Volumes/Hardware
/dev/disk3s5    460Gi  119Gi  318Gi   28%   1.3M  3.3G    0%    /System/Volumes/Data
map auto_home   0Bi    0Bi   0Bi  100%     0    0    -    /System/Volumes/Data/home
/dev/disk5s1    677Mi  525Mi  148Mi   78%   7.9k  1.5M    1%    /Volumes/Arduino IDE 2.3.10-arm64
/dev/disk6s1    1.3Gi  1.3Gi  2.4Mi  100%   11k  4.3G    0%    /Volumes/ChatGPT Installer
amrutha@Amruthas-MacBook-Air ~ %
```

2. du -sh

```
amrutha@Amruthas-MacBook-Air ~ % du -sh .
du: ./Music/Music: Operation not permitted
du: ./Pictures/Photos Library.photoslibrary: Operation not permitted
du: ./Library/Application Support/CallHistoryTransactions: Operation not permitted
du: ./Library/Application Support/CloudDocs: Operation not permitted
du: ./Library/Application Support/com.apple.sharedfilelist: Operation not permitted
du: ./Library/Application Support/Knowledge: Operation not permitted
du: ./Library/Application Support/com.apple.TCC: Operation not permitted
du: ./Library/Application Support/FileProvider: Operation not permitted
du: ./Library/Application Support/AddressBook: Operation not permitted
du: ./Library/Application Support/FaceTime: Operation not permitted
du: ./Library/Application Support/DifferentialPrivacy: Operation not permitted
du: ./Library/Application Support/com.apple.avfoundation/Frecents: Operation not permitted
du: ./Library/Application Support/CallHistoryDB: Operation not permitted
du: ./Library/Assistant/SiriVocabulary: Operation not permitted
du: ./Library/Daemon Containers: Operation not permitted
du: ./Library/Autosave Information: Operation not permitted
du: ./Library/IdentityServices: Operation not permitted
du: ./Library/Messages: Operation not permitted
du: ./Library/HomeKit: Operation not permitted
du: ./Library/Sharing: Operation not permitted
du: ./Library/com.apple.aiml.instrumentation: Operation not permitted
du: ./Library/Mail: Operation not permitted
du: ./Library/Trial: Operation not permitted
du: ./Library/AppleMediaServices: Operation not permitted
du: ./Library/DuetExpertCenter: Operation not permitted
du: ./Library/Accounts: Operation not permitted
du: ./Library/Safari: Operation not permitted
du: ./Library/Bioime: Operation not permitted
du: ./Library/IntelligencePlatform: Operation not permitted
```

3. free -h

```
amrutha@Amruthas-MacBook-Air ~ % top -l 1 | grep PhysMem
PhysMem: 15G used (2709M wired, 5195M compressor), 177M unused.
amrutha@Amruthas-MacBook-Air ~ %
```

Topic 10: User and Environment Management

1. id

```
amrutha@Amruthas-MacBook-Air ~ % id
uid=501(amrutha) gid=20(staff) groups=20(staff),12(everyone),61(localaccounts),79(_appserverusr),80(admin),81(_appserver
adm),33(_appstore),98(_lpadmin),100(_lpoperator),204(_developer),250(_analyticsusers),395(com.apple.access_ftp),398(com.
apple.access_screensharing),399(com.apple.access_ssh),400(com.apple.access_remote_ae),701(com.apple.sharepoint.group.1)
amrutha@Amruthas-MacBook-Air ~ %
```

2. env

```

amrutha@Amruthas-MacBook-Air ~ % env
TMPDIR=/var/folders/h1/gcvz4hvxvnnv15_mqpw_qs0e0000gn/T/
CFBundleIdentifier=com.apple.Terminal
XPC_FLAGS=0x0
TERM=exterm-256color
SSH_AUTH_SOCK=/var/run/com.apple.launchd.WSKLs4dcfr/Listeners
XPC_SERVICE_NAME=0
TERM_PROGRAM=Apple_Terminal
COLORTERM=truecolor
OSLogRateLimit=64
TERM_PROGRAM_VERSION=470.2
TERM_SESSION_ID=862A7243-698C-4EB3-99B4-AF7DMC18D51E
SHELL=/bin/zsh
HOME=/Users/amrutha
LOGNAME=amrutha
USER=amrutha
PATH=/opt/homebrew/opt/postgresql@17/bin:/opt/homebrew/bin:/opt/homebrew/sbin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin
SHLVL=1
PWD=/Users/amrutha
OLDPWD=/Users/amrutha
HOMEBREW_PREFIX=/opt/homebrew
HOMEBREW_CELLAR=/opt/homebrew/Cellar
HOMEBREW_REPOSITORY=/opt/homebrew
PPATH=/opt/homebrew/share/zsh/site-functions:/usr/local/share/zsh/site-functions:/usr/share/zsh/site-functions:/usr/share/zsh/5.9/functions
INFOPATH=/opt/homebrew/share/info:
LANG=en_US.UTF-8
_=/usr/bin/env
amrutha@Amruthas-MacBook-Air ~ %

```

3. export

```

amrutha@Amruthas-MacBook-Air ~ % export MYNAME="Amrutha"
amrutha@Amruthas-MacBook-Air ~ % echo $MYNAME
Amrutha
amrutha@Amruthas-MacBook-Air ~ %

```

4. alias

```

amrutha@Amruthas-MacBook-Air ~ % alias ll='ls -l'
amrutha@Amruthas-MacBook-Air ~ % ll
total 272
-rwxr-xr-x  1 amrutha  staff  33424 Jul 31 13:54 abc
-rw-r--r--  1 amrutha  staff    80 Jul 31 13:51 abc.c
-rw-r--r--  1 amrutha  staff    81 Jul 29 12:39 abc.c#y
drwx-----@ 5 amrutha  staff   160 Jun 19 10:23 Applications
drwxr-xr-x  2 amrutha  staff    64 Aug  7 09:39 blockhack hackathon
drwxr-xr-x  3 amrutha  staff    96 Aug 12 15:31 bookflow_api
drwxr-xr-x  5 amrutha  staff   160 Aug 12 17:20 bookflow_fastapi
-rw-r--r--  1 amrutha  staff  17827 Jul 30 11:02 CREATE
drwxr-xr-x  3 amrutha  staff    96 Aug  3 14:53 data
drwx-----@ 141 amrutha  staff   4512 Aug 12 19:59 Desktop
drwx-----@  5 amrutha  staff   160 Aug 11 13:52 Documents
drwx-----@ 337 amrutha  staff  10784 Aug 12 18:09 Downloads
drwxr-xr-x  3 amrutha  staff    96 Jul 23 12:20 dsa 3 odd sem
-rw-r--r--  1 amrutha  staff    41 Aug 12 19:41 editor_test.txt
-rw-r--r--  2 amrutha  staff    25 Aug 12 20:19 hardlink.txt
-rwxr-xr-x  1 amrutha  staff  33432 Aug 12 19:43 hello
-rw-r--r--  1 amrutha  staff    88 Aug 12 19:43 hello.c
drwx-----@ 91 amrutha  staff   2912 Jul 25 08:50 Library
drwx-----  5 amrutha  staff   160 Apr 22 21:03 Movies
drwx-----+  4 amrutha  staff   128 Mar 20 10:47 Music
-rw-r--r--  1 amrutha  staff    51 Jul 30 11:02 mysql
-rw-r--r--  1 amrutha  staff    39 Aug 12 19:29 names.txt
-rw-r--r--  2 amrutha  staff    25 Aug 12 20:19 original.txt
drwxr-xr-x  3 amrutha  staff    96 Aug 12 18:50 os_test
drwxr-xr-x 17 amrutha  staff   544 Aug  3 12:41 OSSP
-rw-r--r--@  1 amrutha  staff    86 May 11 10:41 package-lock.json
-rwxr-xr-x  1 amrutha  staff     0 Aug 12 20:15 permission.txt
drwx-----+  6 amrutha  staff   192 Jun 13 20:31 Pictures
drwxr-xr-x+  4 amrutha  staff   128 Feb 21 12:34 Public
drwxr-xr-x@  4 amrutha  staff   128 Apr 16 14:18 PyCharmMiscProject
drwxr-xr-x@  3 amrutha  staff    96 Apr 16 14:18 PycharmProjects
-rw-r--r--  1 amrutha  staff     0 Jul 30 11:03 show
lrwxr-xr-x  1 amrutha  staff    12 Aug 12 20:20 symlink.txt -> original.txt
-rw-r--r--  1 amrutha  staff   159 Aug 12 19:07 test.txt
-rw-r--r--  1 amrutha  staff   173 Aug 12 19:28 test2.txt
amrutha@Amruthas-MacBook-Air ~ % █

```